Department of Computer Engineering

Faculty of Engineering, Kasetsart University

HW#4: Design Your Own Database and SQL-II Queries

Student-Designed Database + Student-Formulated Queries

After-Deadline Submissions Will Not Be Accepted

Objective

Design a small relational database of your own choice, create logical sample data, and formulate your own SQL questions. The instructor specifies only the SQL concept that each query must demonstrate. You decide what information is meaningful to retrieve from your database and write both the question and the SQL statement.

SQL-II Scope

- JOIN operators: INNER JOIN, LEFT JOIN, RIGHT JOIN, and OUTER JOIN concepts
- Scalar Subquery in WHERE: =, >, <, >=, <=, <>
- Multi-row Subquery in WHERE: IN, NOT IN, ANY, ALL
- Existence checks: EXISTS and NOT EXISTS
- Review of GROUP BY, aggregate functions, HAVING, and ORDER BY from SQL-I

Part A: Design and Create Your Own Database

Create a database named:

```
hw4_STUDENT_ID
```

Replace STUDENT_ID with your own student ID. Choose any reasonable application domain.

Possible domains: online shopping, food delivery, restaurant, hotel/travel booking, university enrollment, library, hospital, sports, music, or another domain of your choice.

Minimum Database Requirements

- At least 3 related tables.
- Every table must have an appropriate PRIMARY KEY.
- At least 2 FOREIGN KEY constraints must be declared.
- At least one numeric attribute suitable for SUM, AVG, MIN, or MAX.
- At least one unmatched row so that LEFT/RIGHT JOIN or NOT EXISTS can produce a meaningful result.
- Enough logical sample records to support all required SQL concepts (recommended: at least 15-20 rows in total).
- Appropriate MySQL data types and constraints.

Important: You must design data instances that make your own queries meaningful.

Part B: Formulate Your Own SQL Questions => At least 5 Questions are required !!!

For each item, first write a meaningful query question/purpose in English or Thai based on YOUR database. Then write the SQL statement that answers it using the REQUIRED SQL concept.

Example format only: Required concept = INNER JOIN. Your question might be “Show customer names and products they purchased.” Your actual question must fit your own database.

Query 1 - INNER JOIN

Required concept: INNER JOIN

Create a meaningful query combining related rows from at least two tables using INNER JOIN.

Your own query question / purpose:

Query 2 - LEFT JOIN

Required concept: LEFT JOIN

Create a meaningful query preserving ALL rows from the left-side table, including at least one unmatched row.

Your own query question / purpose:

Query 3 - RIGHT JOIN

Required concept: RIGHT JOIN

Create a meaningful query preserving ALL rows from the right-side table, including at least one unmatched row.

Your own query question / purpose:

Query 4 - JOIN + GROUP BY + Aggregate

Required concept: JOIN + GROUP BY + Aggregate

Create a summary query that joins related tables and uses GROUP BY with COUNT, SUM, AVG, MIN, or MAX.

Your own query question / purpose:

Query 5 - Scalar Subquery

Required concept: Scalar Subquery

Use a subquery in WHERE that returns exactly one value and compare it using =, >, <, >=, <=, or <>.

Your own query question / purpose:

Query 6 - IN Subquery

Required concept: IN Subquery

Use IN with a subquery that can return multiple values.

Your own query question / purpose:

Query 7 - NOT IN Subquery

Required concept: NOT IN Subquery

Use NOT IN with a multi-row subquery; your data must make the result meaningful.

Your own query question / purpose:

Query 8 - ANY or ALL Subquery

Required concept: ANY or ALL Subquery

Use either ANY or ALL with a multi-row subquery. State clearly which operator you chose.

Your own query question / purpose:

Query 9 - EXISTS Subquery

Required concept: EXISTS Subquery

Use EXISTS to test whether at least one related row exists.

Your own query question / purpose:

Query 10 - NOT EXISTS Subquery

Required concept: NOT EXISTS Subquery

Use NOT EXISTS to find rows for which no related row exists.

Your own query question / purpose:

Submission Requirements

- Submit ONE file: HW4_Student_ID.sql
- Include CREATE DATABASE, CREATE TABLE, PRIMARY KEY and FOREIGN KEY constraints, INSERT statements, and all 10 queries.
- Before each SQL statement, write your own query question/purpose as a comment.
- Clearly label each query, for example /* Query 1 - INNER JOIN */.
- The complete script must run in MySQL Workbench from beginning to end without manual correction.
- No screenshots or PDF are required unless specifically requested by the instructor or TA.

Required SQL File Format

```
/* Domain: _____________________
   Tables: Table1(...), Table2(...), Table3(...) */

CREATE DATABASE hw4_STUDENT_ID;
USE hw4_STUDENT_ID;

/* Part A: CREATE TABLE + INSERT */
...

/* Query 1 - INNER JOIN
   My question: ____________________________________________ */
SELECT ...;

/* Continue through Query 10 */
```

Checklist Before Submission

☐ I designed my own database domain, schema, and data.

☐ My database has at least 3 related tables with correct PK/FK constraints.

☐ My data supports the required JOIN and subquery situations.

☐ I formulated my own meaningful question for each of the 10 concepts (**at least 5 questions are required**).

☐ Each SQL statement answers the question I wrote and uses the required concept.

☐ The complete .sql script runs successfully in MySQL Workbench.